

AutoML Experiment Report: ECG_AI_fast_checkup

AutoHealth

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Abstract

This report details the development and evaluation of an automated machine learning (AutoML) model for classifying ECG signal health status. The objective was to train a binary classifier to distinguish between *Normal* and *Abnormal* ECG waveforms. The dataset comprised 1D ECG signals with 187 time-point features. A ResNet1D-18 architecture was employed, enhanced with feature selection, Focal Loss to address class imbalance, and Monte Carlo Dropout for uncertainty estimation. The final model achieved a test accuracy of 88.20%, with strong performance on the majority *Abnormal* class (F1: 0.9269) but suboptimal performance on the minority *Normal* class (F1: 0.6935). Uncertainty analysis revealed effective calibration but limited ability to rank predictions by confidence. Key findings include significant class imbalance and potential overfitting, which are the primary targets for subsequent model refinement. All visualizations supporting the analysis are included.

1 Data Analysis

1.1 Dataset Overview

The dataset consists of 1D ECG waveform signals formatted as tabular data, with each sample containing 187 sequential time-point measurements. The data is split into training (2,563 samples), validation (320 samples), and test (322 samples) sets. The target variable is `healthy_status`, with two classes: *Normal* and *Abnormal*.

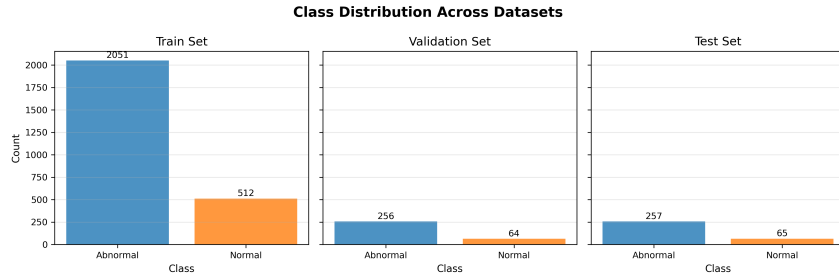


Figure 1: Class distribution across training, validation, and test sets. A significant imbalance is present, with the *Abnormal* class constituting approximately 80% of samples in all splits. This consistency indicates no distributional shift but necessitates strategies to handle the imbalance during model training.

1.2 Feature Analysis and Data Quality

Data quality was high, with no missing values, duplicate rows, or constant columns detected. The waveform values were normalized within the range [0.00, 1.00]. Statistical analysis revealed strong discriminative patterns between classes. For instance, abnormal waveforms exhibited significantly higher mean, standard deviation, and root mean square values, while having lower skewness and kurtosis compared to normal waveforms ($p < 0.001$).

Feature importance analysis identified several time points with high predictive power. The correlation heatmap (Figure 2) shows the top 20 features most correlated with the target. Features 3, 4, and 49-53 showed the strongest positive correlations. The distribution of a key feature (Feature 50) is visualized in Figure 3, demonstrating clear separation between classes and stable distribution across data splits.

2 Model Training

2.1 Data Preprocessing

Preprocessing was designed to enhance model generalization and efficiency. Key steps included:

- **Feature Selection:** To mitigate feature redundancy, the top 100 features with the highest mutual information with the target were selected from the original 187, based on the training set.
- **Standardization:** Selected features were Z-score normalized using statistics from the training set.
- **Class Imbalance Handling:** Focal Loss was employed to increase the focus on the minority (*Normal*) class during training. Hyperparameters for Focal Loss (α and γ) were tuned.
- **Data Augmentation:** To improve robustness, Gaussian noise (with a standard deviation of 0.0045) was applied to 50% of training samples.
- **Calibration Set:** 5% of the training data was held out as an independent set for post-hoc model calibration via temperature scaling.

2.2 Model Architecture

A 1D Residual Network (ResNet1D-18) served as the backbone. To increase model capacity without excessive depth, the initial channel width was increased from 16 to 24. The architecture included Squeeze-and-Excitation (SE) attention modules for channel-wise feature recalibration and a final dropout layer to facilitate Monte Carlo Dropout inference for uncertainty estimation.

2.3 Hyperparameters

A constrained hyperparameter search (4 trials) was conducted, focusing on loss functions and regularization to address class imbalance and overfitting. All trials used the AdamW optimizer, a batch size

Table 1: Hyperparameter Search Trials

Trial	Loss Function	Learning Rate	Dropout	Weight Decay
1	Weighted Cross-Entropy (w=4 for Normal)	1e-3	0.3	1e-4
2	Focal Loss ($\alpha=0.8$, $\gamma=2.0$)	1e-3	0.3	1e-4
3	Focal Loss ($\alpha=0.5$, $\gamma=3.0$)	1e-3	0.3	1e-4
4	Focal Loss ($\alpha=0.8$, $\gamma=2.0$)	5e-4	0.4	5e-4

of 64, the OneCycleLR scheduler, and early stopping with a patience of 5 epochs. Trial 2 (highlighted) achieved the best validation accuracy.

2.4 Training Process

Training was stable across all trials. The best model (Trial 2) converged after 10 epochs before early stopping was triggered. Figure 4 shows the validation accuracy progression for all trials and the loss curves for the best trial. A notable gap between final training accuracy (99.26%) and validation accuracy (91.56%) suggests potential overfitting, indicating a need for stronger regularization in future iterations.

2.5 Model Performance

The final model, calibrated with temperature scaling ($T=0.58$), was evaluated on the held-out test set. Performance metrics are summarized below.

Interpretation & Cautions: The model achieves good overall accuracy (88.20%). However, performance is highly asymmetric. The model excels at identifying *Abnormal* ECGs (Recall: 93.77%, F1: 0.9269) but struggles with the minority *Normal* class (Recall: 66.15%, F1: 0.6935). This is a direct consequence of the 80/20 class imbalance. In a clinical screening context, this means the model is reliable at flagging potential abnormalities but may misclassify a substantial portion of healthy ECGs as abnormal, leading to unnecessary referrals. The confusion matrix (43 True Normal, 22 False Abnormal, 16 False Normal, 241 True Abnormal) confirms this pattern.

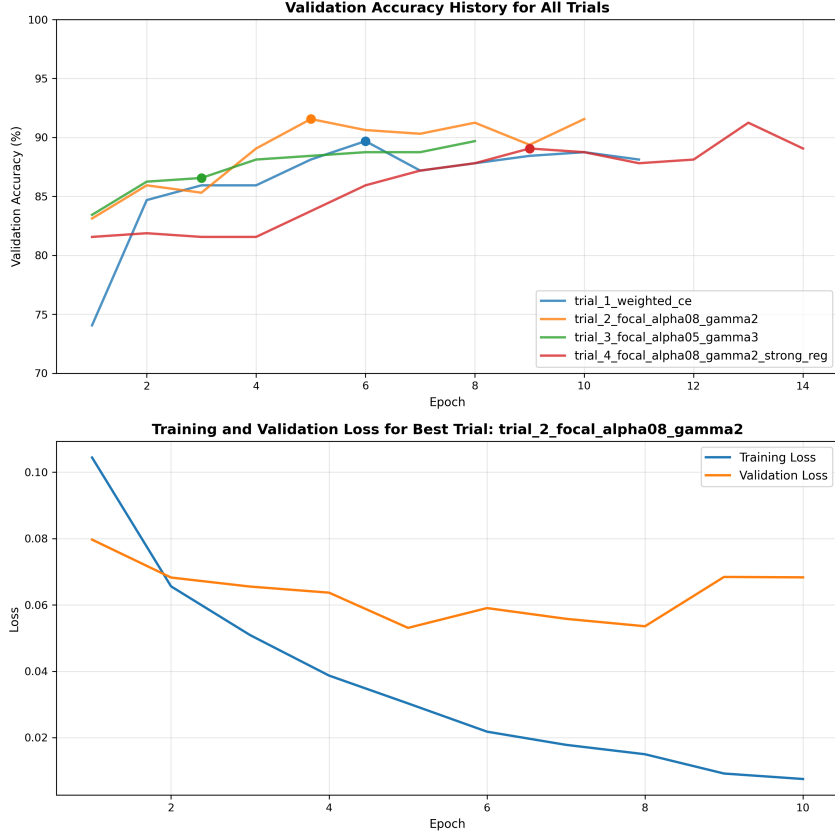


Figure 4: Training history. **(Top)** Validation accuracy across four hyperparameter trials. Trial 2 (Focal Loss with $\alpha=0.8$, $\gamma=2.0$) achieved the highest performance. **(Bottom)** Training and validation loss for the best trial (Trial 2). The sustained gap between training and validation loss, alongside the high final training accuracy, indicates potential overfitting.

Table 2: Final Model Performance on Test Set

Metric	Overall	Normal (Minority)	Abnormal (Majority)
Accuracy	0.8820	-	-
Precision	0.8226 (Macro)	0.7288	0.9163
Recall	0.7996 (Macro)	0.6615	0.9377
F1-Score	0.8102 (Macro)	0.6935	0.9269

3 Uncertainty Analysis

Uncertainty quantification was implemented using Monte Carlo Dropout (20 forward passes) followed by temperature scaling on an independent calibration set.

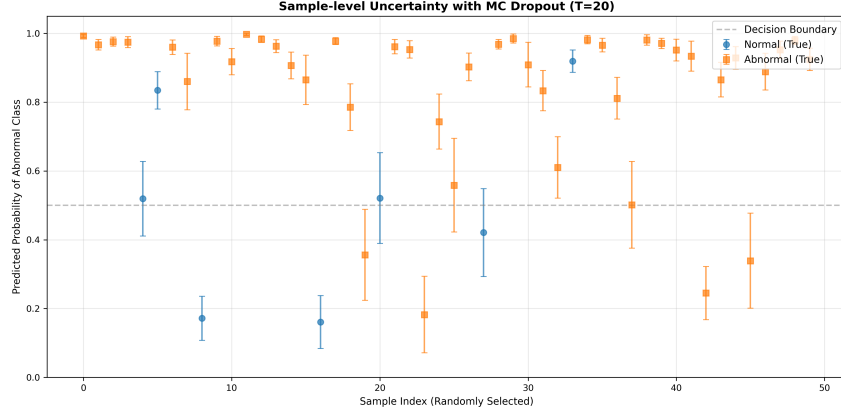


Figure 5: Sample-level predictive uncertainty. Points show the predicted probability of the *Abnormal* class for 50 random test samples, with error bars representing ± 1 standard deviation from MC Dropout. Blue/orange colors indicate true class. Samples near the decision boundary (0.5) and those misclassified often show larger error bars (higher uncertainty), providing a visual cue for low-confidence predictions.

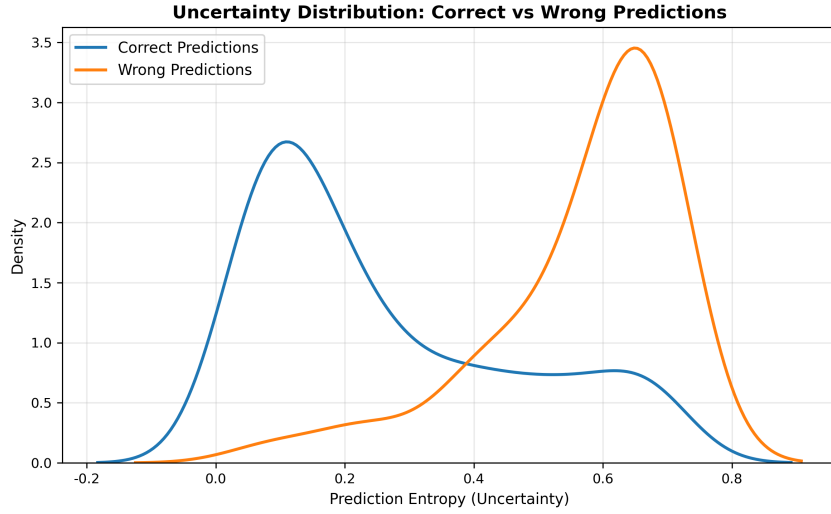


Figure 6: Distribution of predictive entropy for correctly vs. incorrectly classified samples. Incorrect predictions have a significantly higher mean entropy (0.569) than correct predictions (0.260). This indicates the model’s internal uncertainty estimate is **locally effective**—it is generally more uncertain about its mistakes.

Calibration Quality: The model is well-calibrated post-temperature scaling, with a low Expected Calibration Error (ECE) of 0.0357 (Figure 7). This means when the model predicts a 70% probability of abnormality, it is correct approximately 70% of the time, which is crucial for clinical risk assessment.

Uncertainty for Triaging: A key goal is to use uncertainty to identify low-confidence predictions for expert review. Figure 8 shows the Risk-Coverage curve. The negative Area Under the Curve (AUC: -0.0314) indicates that sorting predictions by increasing uncertainty (entropy) does **not** effectively isolate errors. Therefore, while uncertainty is higher for wrong predictions on average (Figure 6), it cannot reliably rank all predictions from most to least trustworthy.

Clinical Guidance:

- **Use Probability for Risk Stratification:** The well-calibrated output probabilities can be used to prioritize cases. A high probability (e.g., > 0.9) of abnormality is very reliable for flagging.

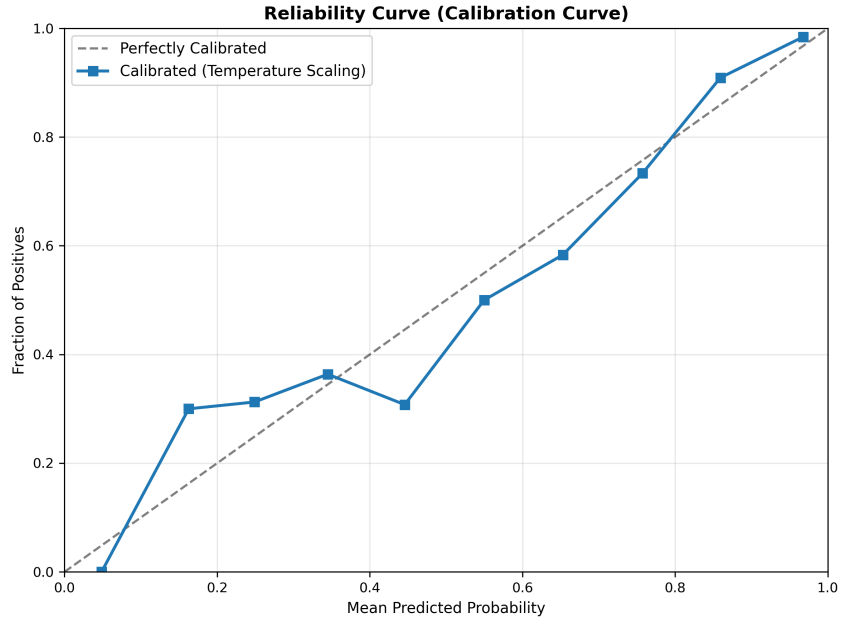


Figure 7: Reliability diagram before and after temperature scaling calibration. The calibrated curve (solid line) closely follows the ideal diagonal, confirming good probability calibration. Deviations in the low-probability range suggest room for further improvement in calibrating low-confidence predictions.

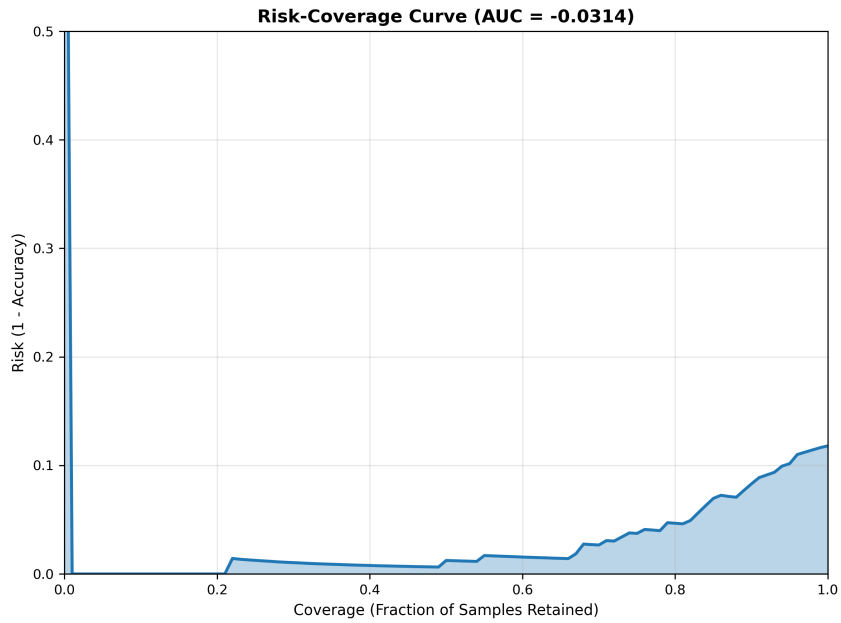


Figure 8: Risk-Coverage curve. The x-axis shows the fraction of test samples retained (coverage) when discarding those with the highest uncertainty. The y-axis shows the error rate (risk) on the retained set. An ideal curve drops sharply. The observed near-linear curve with a negative AUC indicates that the model's uncertainty estimates are not a reliable metric for creating a high-confidence subset.

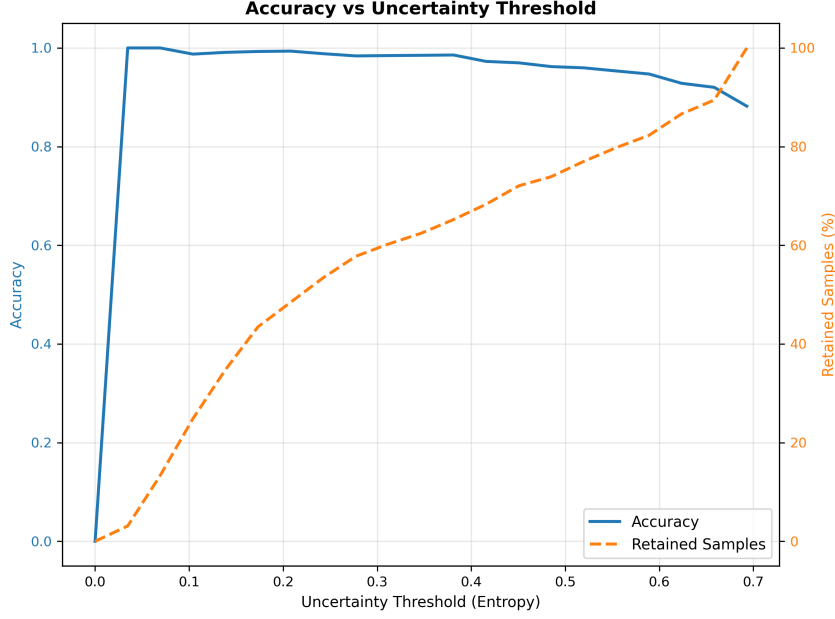


Figure 9: Accuracy and sample retention vs. uncertainty threshold. By rejecting samples with predictive entropy above a threshold, accuracy on the remaining (lower-uncertainty) samples increases. This demonstrates a practical use case: setting an entropy threshold can filter out a portion of predictions, yielding a higher-accuracy subset for automated decision-making, at the cost of coverage.

- **Limit Use of Entropy for Triage:** Do not rely solely on predictive entropy to automatically defer cases for review, as its ranking capability is poor.
- **Employ Threshold-based Filtering:** For a fully automated pipeline requiring very high accuracy, an entropy threshold can be set (e.g., using Figure 9) to filter out uncertain predictions, sending only high-certainty predictions to the output. The trade-off is a reduction in the number of automated decisions.

4 Conclusion

This AutoML pipeline successfully developed a deep learning model for ECG health status classification. The model demonstrates strong overall accuracy (88.20%) and excellent calibration, making its probability outputs clinically interpretable. The primary performance bottleneck is the significant class imbalance, leading to suboptimal recall (66.15%) for the minority *Normal* class. Furthermore, evidence of overfitting (7.70% train-val accuracy gap) and ineffective uncertainty ranking suggest areas for architectural and methodological improvement.

Actionable Recommendations for Next Iteration:

1. **Address Class Imbalance Aggressively:** Increase the class weight for the *Normal* class beyond the tested value of 4.0, or implement sophisticated sampling techniques like SMOTE.
2. **Enhance Regularization:** Increase dropout rates (e.g., to 0.5) and weight decay to mitigate overfitting. Consider simplifying the model architecture.
3. **Improve Uncertainty Ranking:** Experiment with alternative uncertainty quantification methods like Deep Ensembles or explore feature engineering to include signal quality metrics that correlate with model confusion.
4. **Error Analysis:** Manually inspect the 22 False Abnormal and 16 False Normal predictions to identify systematic patterns or data quality issues that can be addressed in preprocessing.

The provided visualizations and uncertainty metrics offer a comprehensive view of model behavior, enabling informed deployment decisions and targeted improvements in subsequent development cycles.